



OPEN Green synthesis of spheroid silver nanoparticles using cyanobacteria with multifunctional biological activities

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Cyanobacteria-mediated synthesis of silver nanoparticles (AgNPs) offers a sustainable approach for producing functional AgNPs for biomedical applications. However, controlling their physicochemical properties while maintaining robust biological activities remains a critical challenge. Here, we developed a green, one-pot synthesis method to produce uniform, small AgNPs using the novel *Cyanobium gracile* PCC 6307 and evaluated their biological activities. PCC 6307 was isolated and identified using microscopic examination and sequencing analysis. The aqueous extract of algal biomass (CB) and cell-free medium (CS) were used to synthesize AgNPs under optimized reaction conditions. Acetylugenol (38.09%) and eugenol (10.5%) were identified as the predominant metabolites in CB extract by gas chromatography-mass spectrometry. CS@AgNPs and CB@AgNPs showed strong absorption peaks at 412.5 and 430.5 nm and exhibited uniform spherical shapes with core diameters of 4.0 and 12.4 nm, respectively. The synthesized AgNPs exhibited hydrodynamic diameters < 100 nm and negative surface charges. Energy-dispersive X-ray spectroscopy confirmed the elemental composition of AgNPs, while Fourier-transform infrared spectroscopy verified the presence of algal functional groups on their surfaces. AgNPs reduced the colon cancer cell viability to 4.1% at 200 µg/mL exhibiting moderate to low toxicity toward normal human fibroblasts. CS@AgNPs revealed higher biocidal activity (inhibition zone ≤ 20 mm) than CB@AgNPs against *Staphylococcus aureus*, *S. haemolyticus*, and *S. lentus*. The antioxidant activity of AgNPs was assessed using the 2,2-diphenyl-1-picrylhydrazyl assay, where AgNPs showed moderate (46.0%) to low (25.1%) radical scavenging activity. This study provides a sustainable approach for producing uniform, small, and biologically active AgNPs with potential biomedical applications.

Keywords Green synthesis, Cyanobacteria, Optimization process, GC-MS

Abbreviations

NPs	nanoparticles
AgNO ₃	silver nitrate
PCR	polymerase chain reaction
nm	nanometer
min	minute

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μL	microliter
mg	milligram
mL	milliliter
μg	microgram
mM	millimolar
h	hour
cm ⁻¹	inverse centimeter
rpm	revolutions per minute
SEM	scanning electron microscopy
TEM	transmission electron microscopy
FTIR	Fourier-transform infrared spectroscopy
kV	kilovolt
UV-vis	ultraviolet-visible spectroscopy
IZ	inhibition zone
DMSO	dimethyl sulfoxide
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-2 H-tetrazolium bromide

Silver nanoparticles (AgNPs) have proven to be highly effective multimodal agents for antimicrobial and cancer therapies due to their distinct physicochemical properties¹. These properties include their small size, high surface-area-to-volume ratio and surface chemistry, which govern their reactivity, stability and functionality². The biological performance of AgNPs is strongly dependent on their size and surface chemistry, which collectively influence the ion release and nano-cell interactions^{3,4}. Consequently, synthesis approaches capable of controlling these parameters are important for effectively integrating AgNPs into biosystems. Physical and chemical synthesis methods are commonly used to produce AgNPs¹. However, these methods are associated with the use of hazardous chemicals and the generation of environmental pollutants. Such chemicals are not only hazardous to the environment but also toxic to normal human cells^{5,6}. To overcome these challenges, new sustainable synthesis approaches are required. Green synthesis offers an eco-friendly alternative for synthesizing AgNPs, where microorganisms or their purified biomolecules, such as proteins, enzymes, and pigments, serve as reducing and stabilizing agents^{7,8}. In green synthesis, no hazardous chemicals are used or produced, and high resource consumption is not required^{9–11}. Among microorganisms, cyanobacteria represent a large and diverse natural source for synthesizing AgNP with various shapes, sizes, and activities. This is likely due to their capacity to produce biomolecules with high redox activity that promote Ag⁺ reduction and stabilize the nanoparticles^{12–14}. Once Ag⁺ reduction occurs, the nucleation and growth step are initiated, where silver atoms deposit on nuclei to grow into particles, followed by stabilization of their surfaces by biomolecules. Factors such as precursor concentration, reaction temperature, and pH can regulate the reaction rate and, consequently, affect the properties of AgNPs^{15,16}. Despite the numerous reports regarding cyanobacteria-mediated synthesis of AgNPs, most studies have not systematically optimized the reaction conditions to modulate the physicochemical properties of the nanoparticles and ultimately the biological performance. Thus, a deep understanding of the optimal factors governing the synthesis of small, stable AgNPs could help address this knowledge gap and facilitate their translation into various biomedical applications. We selected *Cyanobium gracile* PCC 6307, a novel cyanobacterium with a unique metabolite profile, to explore, for the first time, its potential to synthesize AgNPs. *Cyanobium* spp. are rich in biomolecules with high redox potentials, which can promote the reduction and stabilization of AgNPs¹⁷. Therefore, in this study, we used aqueous extracts of *C. gracile* PCC 6307 as reducing and stabilizing agents to produce AgNPs without the need for additional chemicals. This study further advances the scope of biogenic AgNPs by evaluating their in vitro biological activities.

Materials and methods

Materials

Silver nitrate (AgNO₃), chemically synthesized AgNPs (Ch@AgNPs; 99% purity, spherical shape, and nanosize of 100 nm), MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide), and DPPH (2,2-diphenyl-1-picrylhydrazyl) were purchased from Sigma Aldrich (St. Louis, MO, USA). Resazurin dye and all cell culture media supplements were purchased from BDH Chemicals (England) and Gibco (Thermo Fisher Scientific, Waltham, MA, USA), respectively. Ciprofloxacin, nutrient agar, and nutrient broth were obtained from Tokyo Chemical Industry Co., Ltd., Japan, and Himedia, India, respectively. Human skin fibroblasts (HFs) and colon cancer cell lines (Sw620, Sw480, HT-29, and HCT-116) were obtained from the American Type Culture Collection (ATCC, Manassas, VA, USA). Clinical bacterial isolates, including *Staphylococcus aureus*, *S. haemolyticus*, and *S. lentus*, were obtained from the Department of Microbiology, King Saud University, Riyadh, Saudi Arabia.

Methods

Cyanobacteria isolation

Algal samples were collected from freshwater sources in Alexandria, Egypt. The samples were transferred to the laboratory in sterile containers for culture. The samples were first kept in a sterile Petri dish for 7 days to facilitate acclimatization to laboratory conditions (25 °C, fluorescent lamp exposure at 2000 ± 200 lx, 12:12 h dark/light cycle, and BG11 media). The samples were then purified via serial dilution and a zigzag agar-plated assay as described previously^{18,19}. Purified colonies were grown in BG11-containing test tubes for 15 days and subsequently transferred to 250 mL sterile flasks for large-scale culture.

Genetic identification

Genomic DNA was extracted from the algal sample using an Attogene DNA extraction kit (Attogene, Austin, TX, USA) according to the manufacturer's instructions. In brief, the algal cells were collected, mixed with 500 μL of suspension buffer, and vortexed for 1 min. The algal suspension was sonicated in a water bath sonicator at full power for 2 min to break the cells. The algal suspension (200 μL) was transferred to a microcentrifuge tube and mixed with 200 μL of lysis buffer containing poly A. Proteinase K (10 μL) was added, the mixture was vortexed, and incubated at 60 °C for 10 min to digest proteins and release DNA. Following incubation, 400 μL of absolute ethanol was added to the tube to facilitate DNA precipitation. The DNA was collected and washed with washing buffer. The DNA was stored at – 20 °C for further analysis. DNA was amplified using a PeqSTAR 96X universal gradient thermal cycler (VWR, Germany) according to the protocol described by Weisburg et al.²⁰. A forward primer (5'-AGAGTTTGATCCTGGCTCAG-3') and a reverse primer (5'-GGCTACCTTGTTACG ACTT-3') were used for amplification. The amplicons were stored in nuclease-free water and sent to Macrogen (Seoul, Korea) for DNA sequencing.

Morphological analysis

The preliminary taxonomic identification and culture purity of *C. gracile* PCC 6307 were assessed using microscopy. 10 μL of the purified algal suspension was loaded onto a glass slide and examined using light microscopy (Novex, Holland, The Netherlands) and an inverted light microscopy (Zeiss Primovert 5, Germany). Subsequently, a sample was prepared by washing the algal cells three times with distilled H_2O and fixing them in 70% ethanol. 20 μL of this sample was then loaded onto a sterile glass piece attached to a carbon paste on a copper stub, air-dried, and coated with a platinum coater (JEC-3000FC, Joel, Tokyo, Japan) for 80 s for imaging by scanning electron microscopy (SEM, JSM-IT500HR, Joel, Japan) at 15 kV²¹.

Algal extract preparation and phytochemical identification

After 15 days, cyanobacteria cells were harvested by centrifugation at 4700 rpm for 10 min. The biomass was washed with distilled H_2O , freeze-dried in a lyophilizer (LYOTRAP, LTE Scientific, Greenfield, UK) for 24 h, and then ground using glass beads with vortex mixing. 100 mg of algal powder was mixed with 100 mL of distilled H_2O and boiled at 80 °C for 30 min. The mixture was cooled to room temperature, centrifuged at 4700 rpm for 10 min to collect the supernatant, and filtered through filter paper. The cell-free medium was collected and filtered through filter paper. Both filtrates were stored for further applications. Gas chromatography-mass spectrometry (GC-MS) was performed to identify volatile phytochemicals in the aqueous extract of *C. gracile* PCC 6307, as these organic molecules may contribute to AgNP stabilization and influence their physicochemical properties. In brief, 100 mg of algal powder was dissolved in 100 mL of distilled H_2O , heated to boiling, and sonicated for 30 min. Subsequently, the mixture was kept for 24 h, centrifuged, filtered using a syringe filter (0.22 μm), and the filtrate was dried under a vacuum at 50 °C for 48 h. The algal powder was dissolved in methanol and analyzed using a Trace GC-TSQ mass spectrometer (Thermo Scientific, Austin, TX, USA) as described previously²².

Optimization synthesis of AgNPs

AgNPs were synthesized using two methods: cell-free medium and biomass extract-based synthesis. Various parameters were tested, including algal/ AgNO_3 ratio, AgNO_3 concentration, temperature, reaction time, pH, illumination, and incubation time.

Reactant concentration AgNPs were synthesized by mixing 2 mM AgNO_3 with cell-free medium or biomass extract at various ratios (1:1, 1:2 A, 1:4 A, 1:9 A [A represents a higher volume of AgNO_3] and 1:2B, 1:4B, and 1:9B [B represents a higher volume of algal extract]) without pH adjustment. The reaction mixture was maintained at 25 °C under light for 24 h. After 24 h, an aliquot of the AgNPs mixture was analyzed using Ultraviolet-visible (UV-vis) spectroscopy (Shimadzu, Japan). The production of AgNPs with a target size of 10 nm was assessed by an absorption peak at < 430 nm, with blue and red shifts indicating smaller or larger than desired sizes, respectively¹⁶. At the optimum ratios, the algal extracts were mixed with various concentrations of AgNO_3 (1, 2, 4, and 8 mM) under other constant conditions.

Reaction temperature and time The effect of temperature on the size and stability of CS@AgNPs and CB@AgNPs was assessed by exposing the optimum algal/ AgNO_3 mixtures to 40, 60, 80, and 100 °C for 60 min at the initial pH, followed by incubation under light for 24 h. Among AgNPs, CS@AgNP synthesis appeared to be temperature dependent, where small AgNPs formed efficiently at 100 °C. Subsequently, it was necessary to vary the reaction time of cell-free medium/ AgNO_3 mixtures at 100 °C from 60 min to 15 and 30 min while the other conditions were kept constant.

Illumination and incubation time The influence of light on the synthesis of AgNPs was assessed by incubating the algal/ AgNO_3 mixture under light (2000–200 lx) and in the dark, with other conditions kept constant. Subsequently, the influence of the incubation time (24 and 48 h) under light conditions on AgNP synthesis was also assessed. Finally, under constant conditions, the pH of the AgNO_3 and algal supernatant mixture was adjusted using 0.1 M Tris-HCl or NaOH to 6, 7, 8, or 12, while other reactions were kept unadjusted (initial pH 9). Using the optimum parameters, CS@AgNPs and CB@AgNPs were produced on a large scale, centrifuged at 13,400 rpm for 15 min, washed three times with distilled H_2O and freeze-dried for further characterization and biological applications²³.

Characterization of AgNPs

Multiple optical and electronic techniques were used to confirm the formation of AgNPs and evaluate their morphology, size, elemental composition, surface composition, and charge.

TEM, SEM, and EDx The shape, size, and elemental composition of CS@AgNPs and CB@AgNPs were examined using a high-resolution transmission electron microscopy (JEM-2100Flash, Joel, Japan) and SEM combined with a JSMIT500HR energy dispersive X-ray (EDx) detector (STD-PC80, Joel, Japan). For the TEM examination, the AgNPs were collected by centrifugation at 13,400 rpm for 15 min, washed three times with water and ethanol, rinsed in 1 mL of water, and sonicated for 15 min (ultrasonic bath Grant, UK) until a well-dispersed AgNP suspension was obtained. Subsequently, 10 μ L of the AgNP suspension was loaded onto a carbon-coated copper grid (mesh 300) and air-dried. The samples were imaged at 200 kV. The average core diameter of AgNPs was calculated by measuring the diameter of 100 particles using ImageJ (National Institutes of Health, Bethesda, MD, USA). For SEM, 10 μ L of AgNP suspension was loaded onto a glass piece and allowed to dry at room temperature. Other samples were prepared for EDx analysis by loading the AgNP powder onto a carbon paste attached to a copper stub. All samples were then coated with platinum for 80 s and examined at 15 kV¹⁹.

Zetasizer measurements and FTIR The hydrodynamic diameters and surface charges of the AgNPs suspension were measured using a Zetasizer (Malvern, UK). Subsequently, 100 μ L of sample suspension was mixed with 900 μ L of distilled H₂O, sonicated for 15 min, and transferred to Zetasizer tubes at 25 °C for measurement. The surface chemistry of CS@AgNPs and CB@AgNPs powders was examined at 400–4000 cm⁻¹ via FTIR spectroscopy (Shimadzu, Kyoto, Japan).

Anticancer activity

The antiproliferative activities of 200 μ g/mL of CS@AgNPs and CB@AgNPs were assessed against four colon cancer cell lines (HT-29, HCT-116, Sw620, and Sw480) and normal human skin fibroblasts (HFs) using the MTT assay. The samples were prepared by suspending the desired weights of CS@AgNPs and CB@AgNPs in DMEM and sonicated for 15 min. All mixtures were filtered using a 0.45- μ m microfilter, and a serial dilution of AgNPs (200, 100, 50, 25, 12.5, 6.25, 3.13, 1.56, and 0.78 μ g/mL) was prepared. Cells were cultured in their specific RPMI or DMEM media supplemented with 10% FBS, 100 U/mL penicillin and 100 μ g/mL streptomycin and were kept at 37 °C in a 5% CO₂ incubator until 75% confluency was reached. The cells were detached using trypsin-EDTA and centrifuged at 1500 rpm for 5 min. The pellets were suspended in 1 mL of DMEM and counted using a hemocytometer. The cells (5×10^4 cells/well) were then seeded in 96-well plates for 24 h, treated with different concentrations of AgNPs, and incubated for 24 h in a 5% CO₂ incubator at 37 °C. After 24 h of incubation, an MTT assay was performed according to the method described by Mosmann²⁴. The absorbance of samples was measured at 570 nm using BMGLABTECH[®] FLUOstar Omega microplate reader (Ortenberg, Germany).

The cell viability was calculated using the following equation: $(\text{Abs}_{(\text{treated})})/(\text{Abs}_{(\text{control})}) \times 100$.

Antibacterial activity

Staphylococcus aureus, *S. haemolyticus*, and *S. lentus* were cultured in nutrient broth for up to 18 h at 37 °C and maintained by continuous subculturing in broth and on nutrient agar. The biocidal activities of CS@AgNPs, CB@AgNPs, chemically synthesized AgNPs (Ch@AgNPs), AgNO₃, and ciprofloxacin (5 μ g/mL) were assessed using agar well diffusion and microdilution assays. For the agar well diffusion method, *S. aureus*, *S. haemolyticus*, and *S. lentus* were treated with 1000 μ g/mL of the tested chemicals and 5 μ g/mL of ciprofloxacin and incubated for 24 h¹⁹. The inhibition zone (IZ) was measured using a caliper (Series 530, Mitutoyo, Japan).

To determine the minimum inhibitory concentration (MIC) and bactericidal concentration (MBC) of 500 μ g/mL of CS@AgNPs and CB@AgNPs, a microdilution assay using resazurin dye was performed according to the method described by Elshikh et al.²⁵. MBC values were calculated by depositing the contents of wells with concentrations > MIC on nutrient agar plates for 24 h. The MBC refers to the lowest concentration of AgNPs at which no colonies were detected on the plate (zero growth).

Antioxidant activity

The free radical scavenging activity of AgNPs was assessed using the DPPH assay, as described by Sharma et al.²⁶. Briefly, 1000 μ g/mL CS@AgNPs and CB@AgNPs were sonicated for 15 min, and the mixtures were diluted to various concentrations (1000, 500, 250, 125, 62.5, 31.25, 15.6, 7.8, 3.9, and 1.95 μ g/mL). An equal concentration of ascorbic acid was used as a positive control. Subsequently, 100 μ L of each concentration was mixed with 100 μ L of DPPH solution (0.004 g DPPH powder dissolved in 100 mL absolute ethanol) in 96-well plates. Additionally, a blank (ethanol only) and a negative control (DPPH solution only) were prepared. The plates were incubated in the dark for 30 min at ambient temperature. After 30 min, the absorbance of the samples was measured at 517 nm using the xMark™ microplate spectrophotometer (BioRed, USA) and the scavenging activity (%) was calculated using the following equation:

$$\frac{\text{Abs}(\text{control} - \text{blank}) - \text{Abs}(\text{sample} - \text{blank})}{\text{Abs}(\text{control} - \text{blank})} \times 100$$

Statistical section

GraphPad Prism version 9.3.1 (GraphPad Software, Inc., San Diego, CA, USA) and ImageJ were used to analyze and visualize the data. All experiments were independently performed in triplicate and presented as mean $\pm$ SD.

Results and discussion

Morphological appearance of *C. gracile*

After isolation and purification of algal samples, a combination of 16S rRNA sequencing and phylogenetic analysis was performed to identify the novel cyanobacteria and to determine their taxonomic relationship with closely related strains. The current sample was 96.6% identical to *C. gracile* PCC 6307, with a query coverage of 96% (Fig. 1), and its DNA sequence was deposited in GenBank under accession number OP360943.1. The morphology of PCC 6307 was assessed using optical and electron microscopy. The cells appeared small, unicellular, blue-green, coccoid-to-ellipsoid in shape (Fig. 2A–B). Under SEM, PCC 6307 cells had an ellipsoidal shape with a smooth surface (Fig. 2C–D). The morphology of this strain is consistent with previous descriptions of the freshwater *Cyanobium* genus reported by Komárek et al.²⁷ and Kwon et al.²⁸. *C. gracile* was reported to exist as solitary cells lacking enveloping sheaths, but possessing a thin layer of mucilage, with homogeneous pale blue-green cytoplasm and a size of 0.4–2.4 μm $\times$ 0.2–1.0 μm ²⁷.

Phytochemicals analysis of *C. gracile* biomass extract

The volatile organic biomolecules in the PCC 6307 biomass extract were detected using GC-MS (Figure S1 and Table 1). The algal biomass extract contained 21 volatile organic molecules, including bicyclic sesquiterpenes such as caryophyllene-(11); phenylpropanoid compounds such as eugenol and acetyleugenol; macrocyclic organosiloxanes such as tetradecamethylcycloheptasiloxane and hexadecamethylcyclooctasiloxane; fragrance compounds such as cedrenol; fatty acid esters such as hexadecanoic acid, methyl; fatty acid derivatives such as glyceryl palmitate and methyl vaccenate; phytosterols such as beta-sitosterol, and triterpenes such as lup-20(29)-ene-3,28-diol. The presence of these diverse organic molecules highlights the potential of PCC 6307 as a promising source of bioactive metabolites. Among these metabolites, acetyleugenol (38.09%) and eugenol (10.5%) were identified as the predominant metabolites based on peak area (%). The high abundance of these phenolic compounds with strong redox potential suggests a critical role in AgNP synthesis. Our results agree with previous reports where *Cyanobium* spp. LEGE 06113 displayed high contents of carotenoids and phycobiliproteins with bioactive properties due to their high anti-hyaluronidase potential²⁹.

Optimization of AgNP synthesis

Algal/AgNO₃ ratio, AgNO₃ concentration, temperature, reaction time, pH, illumination, and incubation time were modulated to achieve the desired size and stability of AgNPs (Fig. 3). The synthesis of AgNPs with the desired size (~10 nm) was indicated by a plasmonic resonance peak at ~430 nm, with blue and red shifts indicating the formation of smaller and larger AgNPs, respectively¹⁶. The optimum algal/AgNO₃ ratios were assessed as 1:9B and 1:4A for CS@AgNPs and CB@AgNPs, respectively, with the corresponding absorption peaks observed at 423.5 nm and 430.5 nm. Increasing the concentration of AgNO₃ above 2 and 4 mM enhanced the formation of larger AgNPs or their aggregation, as confirmed by red shifts in their absorption peaks at 423.5 and 430.5 nm, respectively. For CS@AgNPs and CB@AgNPs, the optimal reaction temperatures were 100 °C and 25 °C, respectively. The formation of CS@AgNPs appeared strongly temperature-dependent, where the organic molecules responsible for reducing AgNPs may reach an optimal level of activation at 100 °C, leading to an increase in the nucleation kinetics constant rather than a decrease in the growth kinetics, resulting in the formation of small AgNPs³⁰. On the contrary, increasing the reaction temperature resulted in the formation of larger or unstable CB@AgNPs. This could be attributed to (i) Ostwald ripening, where smaller particles dissolve and redeposit onto larger particles, a process accelerated by higher temperatures, leading to a broader size distribution and aggregation^{31,32}. (ii) Degradation and denaturation of algal biomolecules, which are responsible for the stabilization of AgNPs under high temperature. The optimal reaction time for producing CS@AgNPs was 60 min, with an absorption peak at 412.5 nm. Light exposure was required for efficient AgNP synthesis, suggesting that light may activate organic molecules that reduce and stabilize AgNPs¹⁹. Under dark conditions, the absorption peaks of AgNPs were broad, and their wavelengths were red-shifted, indicating the formation of unstable and larger AgNPs. Increasing the incubation time of AgNO₃/algal extract mixtures under light from 24 h to 48 h, resulting in red-shifts in their absorption peaks from 412.5 to 430.5 nm to 421.5 and 441 nm, respectively, highlighting the formation of larger AgNPs or aggregates. The optimal pH for producing CS@AgNPs and CB@AgNPs was 9 and 7, respectively. Our results agree with previous reports showing that, at higher pH (> 7), functional groups become deprotonated, thereby strongly enhancing electrostatic interactions between negatively charged functional groups, such as carboxyl groups, and the positively charged silver ions, favoring electrostatic repulsion between nanoparticles and discouraging aggregation^{33,34}. Overall, the optimum conditions for synthesizing CS@AgNPs and CB@AgNPs were 2 mM AgNO₃, 1:9B, 100 °C for 60 min, under light incubation for 24 h at pH 9; and 4 mM AgNO₃, 1:4A, 25 °C, under light incubation for 24 h at pH 7; respectively (Fig. 4A).

Characterization of AgNPs

The AgNPs produced under optimized conditions were characterized by TEM, SEM, EDx, Zetasizer measurements, and FTIR. CS@AgNPs and CB@AgNPs were mainly composed of uniform spherical particles with lattice plane distances of 0.23 and 0.24 nm, respectively (Fig. 4B and Figure S2). The average core diameters of CS@AgNPs and CB@AgNPs were 4.0 $\pm$ 0.1 nm and 12.4 $\pm$ 0.6 nm, respectively (Figure S2). Unlike TEM, the resolution of SEM was insufficient to clearly resolve the morphology of AgNPs. However, the predominant morphologies of AgNPs were spherical, where CS@AgNPs being uniformly distributed, whereas CB@AgNPs formed clusters (Figure S3). Our data agree with previous reports where *Synechocystis* sp. were able to synthesize spherical AgNPs with an average core diameter of 16 nm larger than that produced by our method³⁵. The elemental composition of CS@AgNPs and CB@AgNPs was evaluated by EDx analysis. EDx signals of Ag at 3 keV were observed, where silver represented 70.45 $\pm$ 0.11% and 75.65 $\pm$ 0.11%, respectively (Fig. 5 and Table S1).

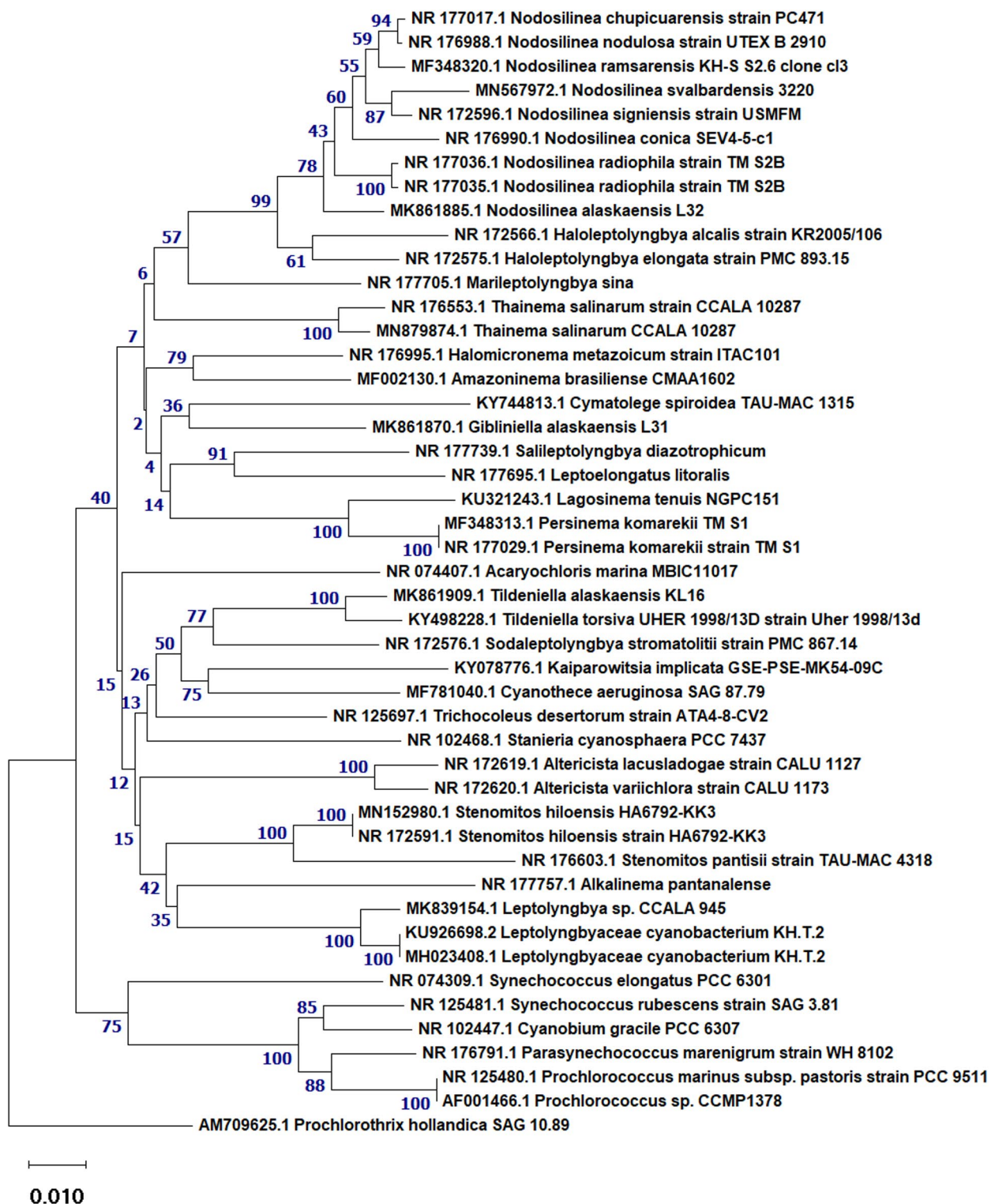


Fig. 1. Phylogenetic tree of *Cyanobium gracile* PCC 6307 inferred from 16S rRNA sequences and constructed using MEGAX software (version 10.2.6). The numbers at the branch points represent bootstrap values (%) based on 1000 replicates generated using the neighbor-joining method.

Carbon was the second-most abundant element in CS@AgNPs and CB@AgNPs with a mass % of $12.44 \pm 0.02\%$ and $11.90 \pm 0.02\%$, respectively. Trace amounts of oxygen, phosphorus, chloride, copper, and zinc were observed. These elements are the main components of BG11 medium used for algal growth. Therefore, the presence of these elements in AgNPs may be associated with organic molecules derived from the algal extract, which act as reducing and stabilizing agents during AgNP synthesis. AgNPs synthesized using the cell biomass extract of

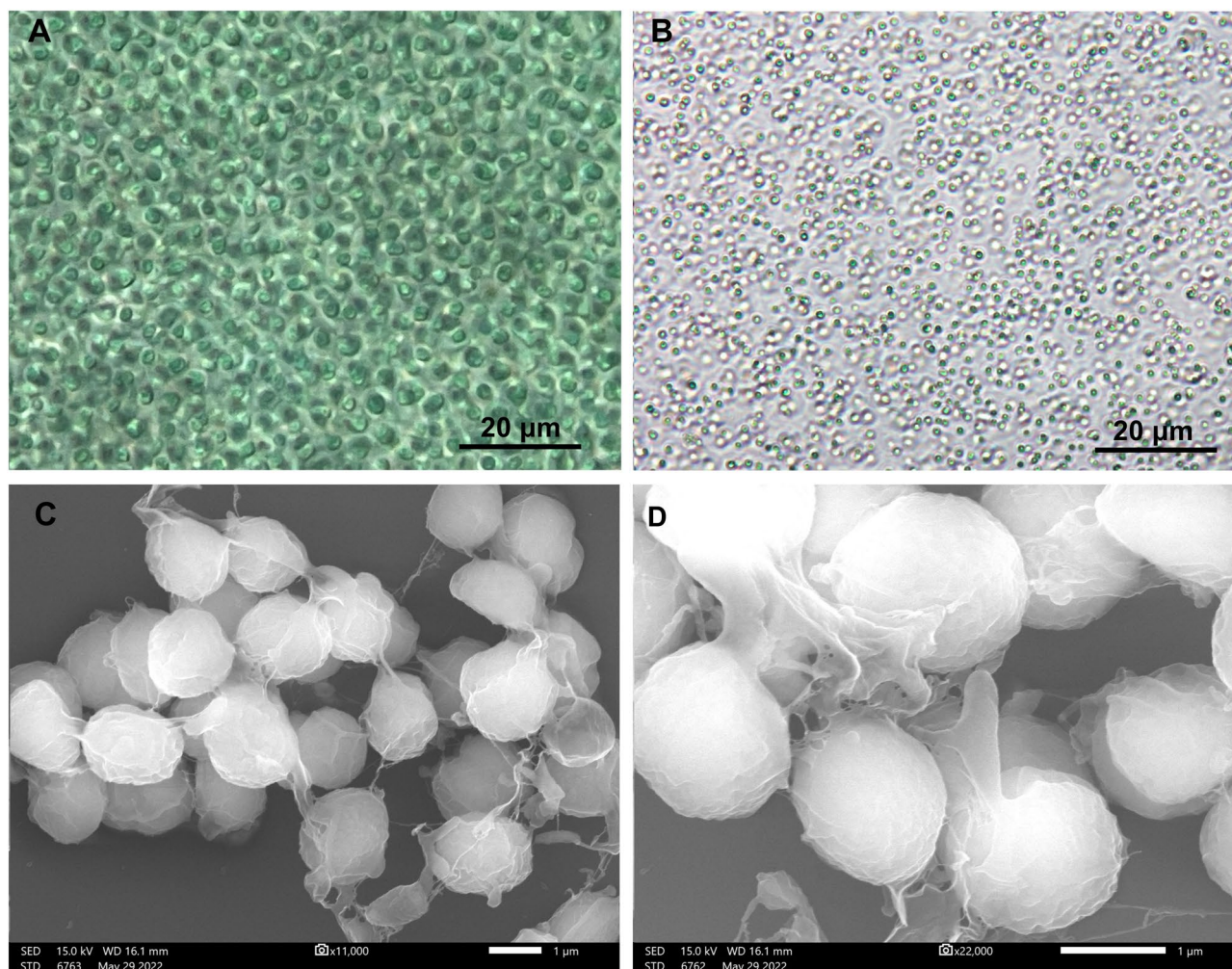
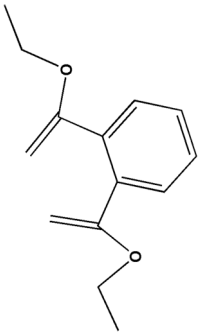
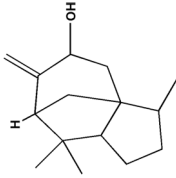
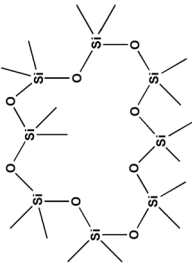
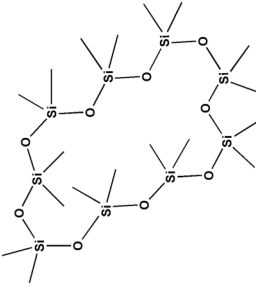


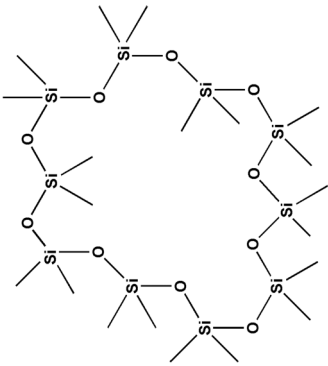
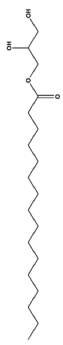
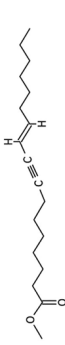
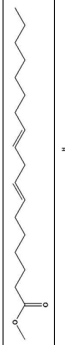
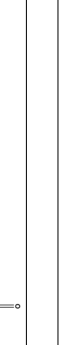
Fig. 2. Light microscopy (A), inverted light microscopy (B) and scanning electron microscopy (C–D) images illustrating the morphological appearance of *Cyanobium gracile* PCC 6307. Cells appeared as small, unicellular, blue-green coccoid to ellipsoid in shape with smooth surfaces. Scale bar, 20 μm (A–B) and (C–D) 1 μm .

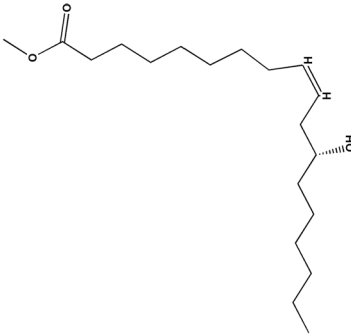
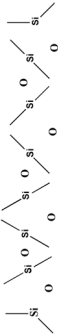
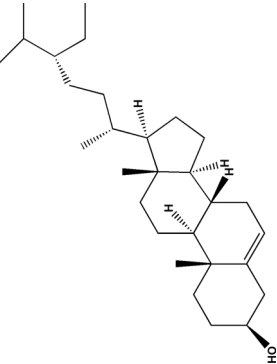
Leptolyngbyasp. WUC 59 showed a clear signal at 3 keV, whereas other low-intensity peaks correspond to algal biomolecules bound to the AgNP surfaces³⁶.

Zetasizer measurements were performed to evaluate the hydrodynamic diameters (HDs) and surface charges of AgNPs in aqueous media. The results showed that CB@AgNPs had smaller HDs (25.44 ± 7.7 nm) while CS@AgNPs had larger HDs (97.05 ± 76.0 nm), with PDI values of 0.26 and 0.396, respectively (Fig. 4C and Figure S4). Despite TEM analysis showing that AgNPs had small core diameters, the HDs were larger. This is likely due to fundamental differences between the two techniques. TEM measures the core diameters of AgNPs in their dry state. At the same time, Dynamic Light Scattering measurements report the size of AgNPs in solutions, including the solvating and associated hydrophilic layers (algal corona) surrounding AgNPs. Therefore, the high HD values of our biogenic AgNPs could be attributed to the presence of algal biomolecules, such as proteins and polysaccharides, coating the AgNPs and to their affinity for water molecules. Although CB@AgNPs have an algal corona as detected using FTIR (Fig. 4E), their HDs were smaller than those of CS@AgNPs. This could be attributed to differences in the thickness of the algal corona and the composition of the algal corona-coated AgNPs. CS@AgNPs and CB@AgNPs had -23.6 ± 7.2 mV and -12.8 ± 11.3 mV, respectively (Fig. 4D and Figure S3). The negative surface charges of AgNPs may be due to the presence of negative functional groups on AgNP surfaces derived from cyanobacteria³⁶. The variation in z-potential values of resultant AgNPs may indicate differences in their surface chemistry. Our results are consistent with previous reports in which AgNPs synthesized using cell-free extract of *Chlorella vulgaris* had a z-potential of -26.4 mV, whereas AgNPs synthesized by exopolysaccharides (EPS) of *Chlorella pyrenoidosa* and *Botryococcus braunii* exhibited z-potential of -12.16 ± 2.41 and -51.81 ± 3.01 mV, respectively^{37,38}. Gallón et al. reported that the negative charge on AgNPs synthesized by EPS indicated that anionic EPS coated them. These EPS enhance the stabilization of AgNP not only through electrostatic interactions but also through steric effects via hydrogen bonds within the EPS matrix, which coat AgNP surfaces and prevent their agglomeration³⁷.

No.	Compound	Retention time	Area%	Matched factor	Molecular formula	Molecular weight	Chemical structure
1	Eugenol	13.21, 13.33	5.69, 4.81 (10.5)	948, 944	$C_{10}H_{12}O_2$	164	
2	Caryophyllene-(II)	14.92	0.93	914	$C_{15}H_{24}$	204	
3	Tetradecamethylcyclodeptasiloxane	16.24	1.06	890	$C_{14}H_{42}O_7Si_7$	519	
4	Acetyleugenol	17.20, 17.28	19.72, 18.37 (38.09)	953, 916	$C_{12}H_{14}O_3$	206	
5	Caryophyllene oxide	18.71, 18.79	0.30, 1.15	816, 824	$C_{15}H_{24}O$	220	
Continued							

No.	Compound	Retention time	Area%	Matched factor	Molecular formula	Molecular weight	Chemical structure
6	Diethyl Phthalate	18.93	0.74	843	$C_{12}H_{14}O_4$	222	
7	Cedrenol	19.40	0.30	794	$C_{13}H_{24}O$	220	
8	Hexadecamethylcyclononasiloxane	20.03	2.62	927	$C_{16}H_{48}O_8Si_8$	593	
9	Octadecamethylcyclononasiloxane	23.34, 31.47	2.81, 2.86	863, 828	$C_{18}H_{54}O_9Si_9$	667	
10	Hexadecanoic acid, methyl ester	26.16	1.59	903	$C_{17}H_{34}O_2$	270	
Continued							

No.	Compound	Retention time	Area%	Matched factor	Molecular formula	Molecular weight	Chemical structure
11	Cyclodecasiloxane, eicosamethyl-	26.30, 29.00	3.04, 2.81	806, 751	$C_{20}H_{60}O_{10}Si_{10}$	741	
12	1-Monopalmitic n	27.14	0.99	751	$C_{19}H_{38}O_2$	330	
13	10-Heptadecen-8-ynoic acid, methyl ester, (E)-	27.24	0.69	769	$C_{18}H_{30}O_2$	278	
14	7,10-Octadecadienoic acid, methyl ester	29.31	1.28	876	$C_{19}H_{34}O_2$	294	
15	Methyl vaccenate	29.44	1.72	870	$C_{19}H_{36}O_2$	296	
16	Methyl isostearate	29.96	0.48	840	$C_{19}H_{38}O_2$	298	
Continued							

No.	Compound	Retention time	Area%	Matched factor	Molecular formula	Molecular weight	Chemical structure
17	Methyl ricinoleate	32.70, 32.79	0.65, 0.93	784, 845	$C_{19}H_{36}O_3$	312	
18	1,1,3,3,5,5,7,7,9,9,11,11,13,13,15,15-Hexadecamethyloctasiloxane	33.75, 35.91, 37.94, 39.86	2.88, 2.53, 2.46, 1.27	829, 795, 802, 804	$C_{16}H_{50}O_8Si_8$	579	
19	beta-sitosterol	35.55	4.34	882	$C_{29}H_{50}O$	414	
Continued							

No.	Compound	Retention time	Area%	Matched factor	Molecular formula	Molecular weight	Chemical structure
20	Di(2-ethylhexyl) phthalate	36.62	1.43	779	C ₂₄ H ₃₈ O ₄	390	
21	Lup-20(29)-ene-3,28-diol	40.0	2.78	866	C ₃₀ H ₅₀ O ₂	442	

Table 1. Volatile phytochemicals detected in *Cyanobium gracile* extract using GC-MS.

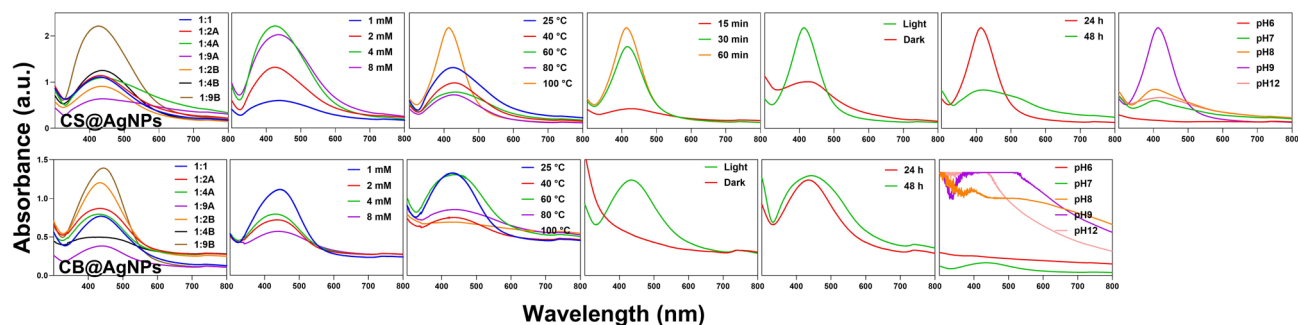


Fig. 3. UV-vis absorption spectra of CS@AgNPs (Top) and CB@AgNPs (Bottom) synthesized under various optimized reaction parameters, including algal/AgNO₃ ratios (V/V, A indicates a higher volume of precursor, whereas B indicates a higher volume of algal supernatant), AgNO₃ concentrations, reaction temperature, reaction time, illumination conditions, incubation time under light, and pH.

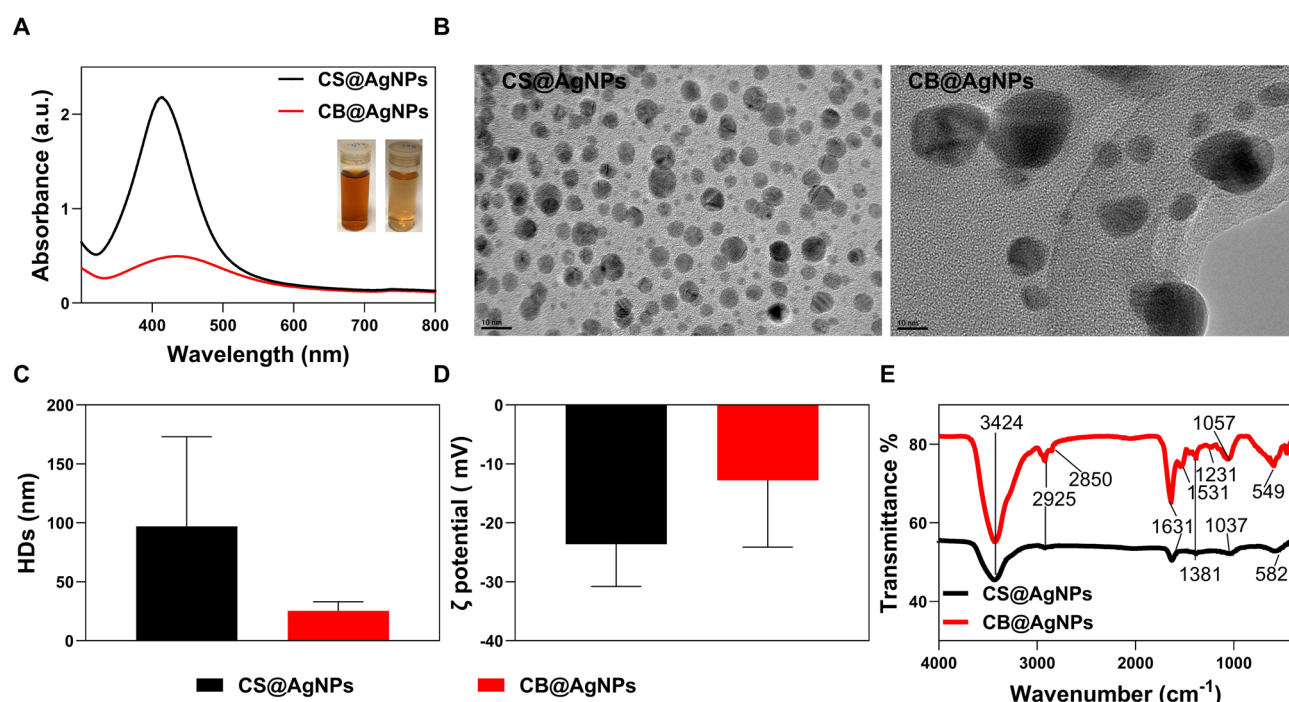


Fig. 4. Physicochemical characterization of CS@AgNPs and CB@AgNPs synthesized under optimal reaction conditions. **(A)** UV-vis absorption spectra. **(B)** Morphologies of AgNPs observed using a transmission electron microscope. Scale bar, 10 nm. **(C)** Hydrodynamic diameter distribution (nm) and **(D)** Zeta potential (mV) of AgNPs. **(E)** Fourier transform Infrared spectra (FTIR) illustrating the functional groups associated with AgNP surfaces.

FTIR analysis was performed to assess the functional groups on the surfaces of CS@AgNP and CB@AgNP, as shown in Fig. 4E. The main functional groups detected in CS@AgNPs were O-H, C-H, C=C, C-O (alkyl, aryl and ether), and C-I, whereas those in CB@AgNPs were O-H, C-H, C=C, N-O, C-O (alcohol, alkyl, aryl, and ether), and C-I. Both CS@AgNPs and CB@AgNPs shared common functional groups, such as O-H, C-H, and C=C; however, CB@AgNPs contained C-O groups from alcohols and N-O from nitro compounds, which were absent in CS@AgNPs. These functional groups are derived from algal biomolecules, such as polysaccharides, proteins, secondary metabolites, phenolic compounds, or fatty acids, which play an important role in reducing and stabilizing AgNPs. Although the exact mechanism of cyanobacteria-mediated synthesis of AgNPs remains under investigation, many reports have demonstrated that the redox activity of phytochemicals in algal extracts is the primary factor in reducing and stabilizing AgNPs¹³. Algal extracts contain a variety of biomolecules with high radical-scavenging activity. These biomolecules can donate electrons to Ag⁺, reducing them to Ag⁰ and thereby activating nucleation and growth steps. During this process, these biomolecules bind to the AgNP surface, forming a corona that stabilizes the AgNPs. Based on this, we propose that the reduction and stabilization of AgNPs are strongly influenced by organic molecules identified by GC-MS analysis (Figure S1

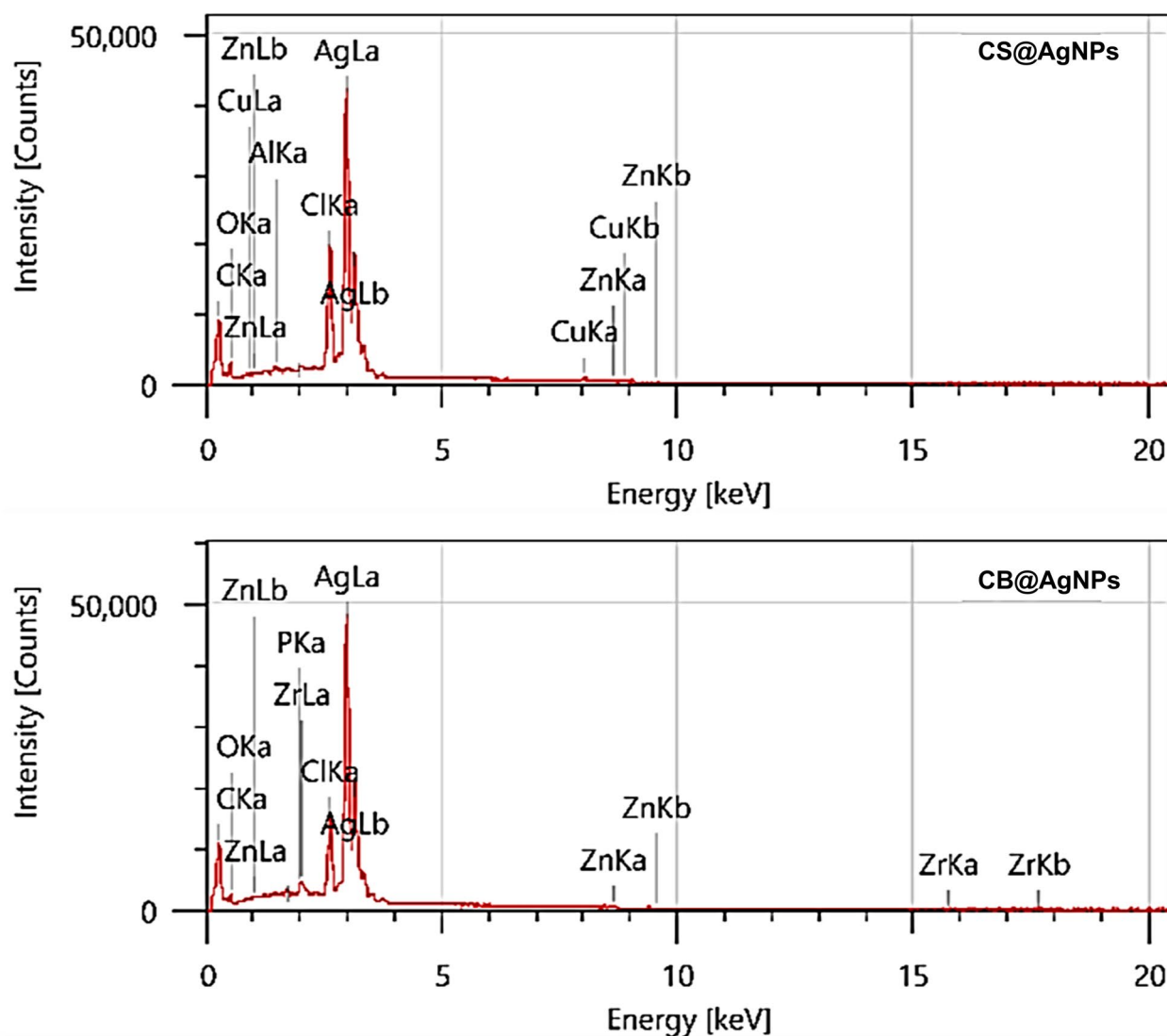


Fig. 5. EDx spectra illustrating the elemental compositions of CS@AgNPs (Top) and CB@AgNPs (Bottom).

and Table 1). Phenolic compounds such as eugenol are well known for their redox activity³⁹, enabling electron donation to Ag^+ and thereby facilitating the reduction process of these ions. In addition to their capacity to promote reduction, these biomolecules could adsorb onto the AgNP surface, providing steric stability. The slight difference in chemical composition between CS@AgNPs and CB@AgNPs suggests that the synthesis mechanism may vary slightly depending on the extraction method and biomolecule concentration. Our results align with previous reports, showing that proteins, alcohols, and phenols in *Leptolyngbya* sp. extract have a dual function: reducing and stabilizing AgNPs³⁶.

Anticancer activity of AgNPs

The cytotoxicity of CS@AgNPs and CB@AgNPs against the colon cancer cell lines HT-29, HCT-116, Sw620, and Sw480, as well as the normal human fibroblast (HFs) cell line, was assessed using the MTT assay (Fig. 6). Both nanostructures exhibited dose-dependent reductions in cell viability. CB@AgNPs showed significantly higher cytotoxicity than CS@AgNPs across all tested cells, reducing cell viability to 4.1–7.8% at 200 $\mu\text{g/mL}$, whereas CS@AgNPs caused ~50% inhibition in cell viability under the same concentration^{40,41}. This enhanced anticancer activity may be explained by the combined effects of particle size and algal corona composition. Despite CB@AgNPs being larger (12.4 nm), they fall within the size range of clathrin- and caveolae-mediated endocytosis, facilitating their cellular internalization⁴². Although ultrasmall CS@AgNPs (4 nm) are highly reactive, they may exhibit rapid, nonspecific interactions, that limit their internalization and reduce their toxicity. Our data agree with a previous report where larger NPs (40–50 nm) can act as crosslinking agents, promote receptor clustering, enhance uptake, and enable membrane wrapping⁴⁰. Algal corona is composed of active phytochemicals, which play an important role in regulating cell/NP interactions by enhancing their stability and facilitating their cellular internalization. Thus, this corona may contribute to cytotoxicity by disrupting the redox balance and

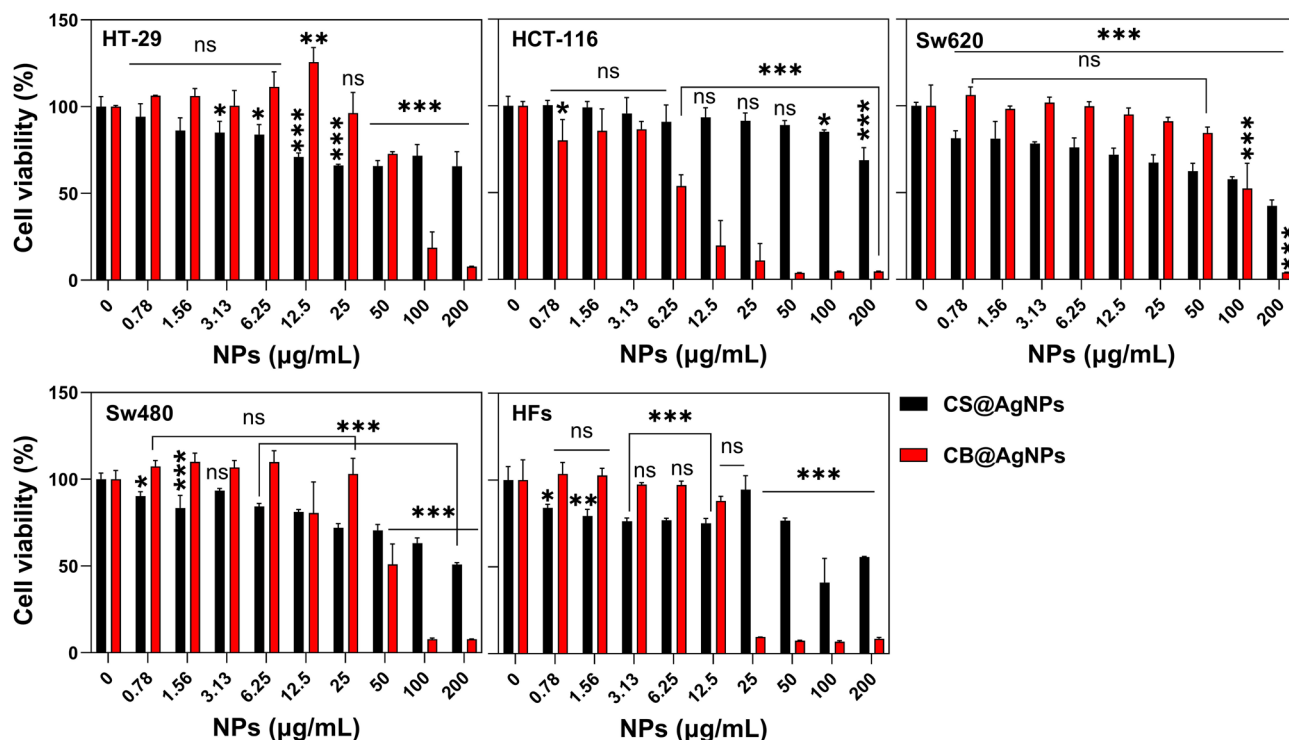


Fig. 6. HT-29, HCT-116, SW620, SW480 and HF cell viabilities after 24 h of incubation with various concentrations of CS@AgNPs and CB@AgNPs. Data are presented as mean $\pm$ SD. p-values were calculated versus untreated cells using one-way analysis of variance (ANOVA) followed by Tukey's test: *** $p < 0.0001$, ** $p < 0.001$, and * $p < 0.01$, ns, not significant ($n = 3$).

increasing oxidative stress. Such mechanisms align with previous reports on algae-mediated synthesis of AgNPs, in which pronounced antiproliferative activity has been attributed to the potential of biogenic AgNPs to induce the formation of reactive oxygen species (ROS)^{43,44}.

Among the tested cell lines, HCT-116 cells were the most sensitive to CB@AgNPs, with an IC_{50} of 6 μ g/mL (Table S2). This could be due to intrinsic biological characteristics of HCT-116, including their metabolic state, cellular mechanisms, and cell resistance. Our data agree with a previous report showing that AgNPs synthesized by *Oscillatoria limnetica* exhibited significant anticancer activity against HCT-116 and MCF-7 cells, with IC_{50} values of 5.369 and 6.147 μ g/mL, respectively⁴⁵. Despite the anticancer activity of CB@AgNPs, their toxicity toward HF cells was greater than that of CS@AgNPs, suggesting a limitation in selectivity. This lack of selectivity between malignant and normal cells indicates that the enhanced bioactivity conferred by the algal corona may also promote nonspecific interactions. Accordingly, while CB@AgNPs exhibit notable anticancer activity, further studies are required to understand their corona composition to improve cancer selectivity and minimize side effects. CB@AgNPs showed greater anticancer activity than Ch@AgNPs (Table S3)¹⁹, highlighting the significant role of the algal corona bound to AgNPs in enhancing their anticancer efficacy. Furthermore, CB@AgNPs exhibited greater anticancer efficacy at lower concentrations against the tested malignant cells than 5-FU¹⁹. However, it should be noted that this study was limited to in vitro investigations and did not examine the underlying molecular mechanisms, particularly those associated with ROS generation.

Antioxidant activity of AgNPs

The DPPH assay was performed to evaluate the antioxidant activity of the resultant AgNPs. The radicals scavenging activity of CB@AgNPs ($46.0 \pm 2.6\%$) was higher than that of CS@AgNPs ($25.1 \pm 0.2\%$) at 1000 μ g/mL but lower than that of ascorbic acid ($82.7 \pm 0.8\%$) (Fig. 7A). The lower antioxidant activity of biogenic AgNPs could be attributed to the chemical composition of these AgNPs, in which phytochemicals bind to AgNP surfaces but possess low redox potential or are present at low concentrations. AgNPs synthesized extracellularly by *Microchaete* showed higher radicals scavenging activity ($80.3 \pm 2.4\%$) than the algal extract ($10.0 \pm 1.1\%$) and was similar to that of ascorbic acid ($95.0 \pm 1.4\%$)⁴⁶.

Antibacterial activity of AgNPs

Biogenic AgNPs exhibited potent inhibitory activity against *S. aureus*, *S. haemolyticus*, and *S. lentus* (Figure S5). The MIC and MBC values of AgNPs required to inhibit and kill bacteria ranged from 125 to 250 μ g/mL (Table 2). CS@AgNPs showed greater antibacterial potency against the tested microbes than CB@AgNPs. The higher antibacterial activity of CS@AgNPs could be due to their small size and large surface area, leading to the release of more silver ions from the AgNP surface, which subsequently bind to bacterial membranes via

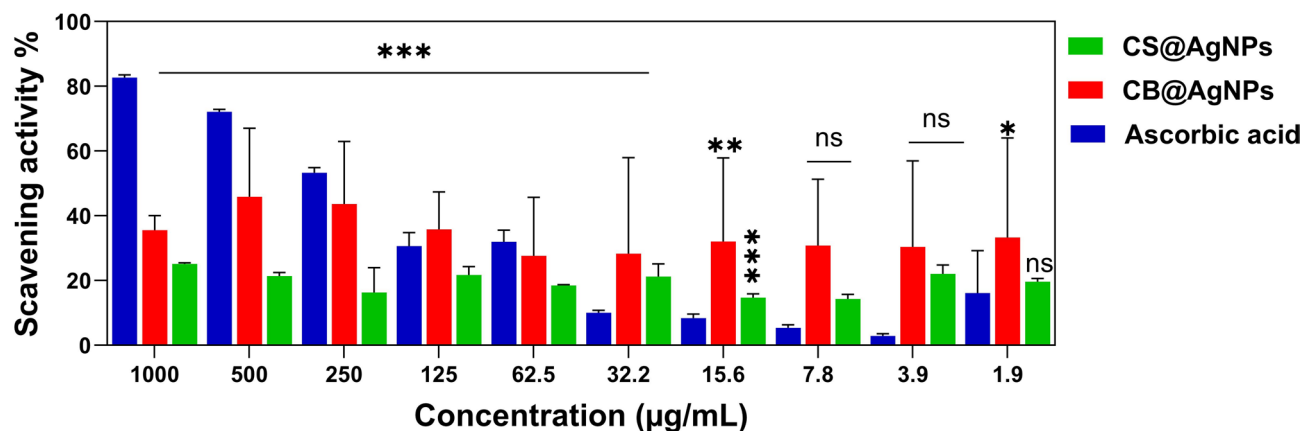


Fig. 7. Radical scavenging activity (%) of CS@AgNPs and CB@AgNPs at 1000 µg/mL. Data are presented as mean $\pm$ SD. p-values were calculated versus ascorbic acid (positive control) using two-way ANOVA followed by Dunnett's test: *** $p < 0.0001$, ** $p < 0.001$, and * $p < 0.01$, ns, not significant ($n = 3$).

Drugs/Bacteria	Staphylococcus aureus		Staphylococcus haemolyticus		Staphylococcus lentus	
	MIC	MBC	MIC	MBC	MIC	MBC
CS@AgNPs	31.25	62.5	125	250	62.5	125
CB@AgNPs	62.5	125	125	250	125	250
Inhibition zone (mm)						
CS@AgNPs	20.8 $\pm$ 0.05		18.2 $\pm$ 0.1		17.1 $\pm$ 0.1	
CB@AgNPs	19.7 $\pm$ 0.1		17.8 $\pm$ 0.1		17.3 $\pm$ 0.1	
AgNO ₃	17.0 $\pm$ 0.03		15.1 $\pm$ 0.19		15.0 $\pm$ 0.13	
Ch@Ag-NPs	16.6 $\pm$ 0.03		13.8 $\pm$ 0.03		13.0 $\pm$ 0.05	
Ciprofloxacin	38.8 $\pm$ 0.1		40.0 $\pm$ 0.5		38.3 $\pm$ 0.6	

Table 2. Antibacterial activity of ciprofloxacin, CS@AgNPs, CB@AgNPs, AgNO₃ and Ch@AgNPs against *Staphylococcus aureus*, *S. haemolyticus*, and *S. lentus*.

electrostatic interactions with sulfur-containing proteins. This may increase membrane permeability, leading to bacterial membrane rupture^{47–49}. Furthermore, AgNPs (< 10 nm) can easily penetrate the bacterial cell wall and interact with DNA, proteins, lipids, and, in some cases, respiratory enzymes. This may lead to ROS production, exacerbating oxidative stress and causing nucleic acid and protein damage, ultimately leading to cell death⁵⁰. Our results agree with previous reports where AgNPs synthesized by *Streptomyces enissocaesilis* caused bacterial membrane deformation in *S. aureus*, *E. coli*, and *Pseudomonas aeruginosa*⁵¹. These membrane deformations were attributed to the small size and the spherical shape of AgNPs, which present a larger surface area, facilitating greater interactions with bacteria.

The high potency of CS@AgNPs and CB@AgNPs against *Staphylococcus sp.* compared to AgNO₃ and Ch@AgNPs could be attributed to the algal corona bound to AgNPs, providing good colloidal stability and consequently enhancing their biocidal activity. These functional groups may contribute to enhanced interactions between AgNPs and bacterial cell membranes, thereby facilitating the cellular uptake of AgNPs. In addition, the smaller size of AgNPs facilitates their penetration of bacterial cell membranes, and their large surface area enables greater contact with the bacterial membrane, causing greater membrane damage and bacterial death^{10,52}. Thyme-mediated AgNPs represent a potent synergistic antimicrobial agent against methicillin-resistant *S. aureus*, indicating the critical role of functional groups in enhancing antibacterial activity of AgNPs⁵³. Among the tested *Staphylococcus* species, *S. aureus* was the most sensitive to all treatments, followed by *S. haemolyticus* and *S. lentus*. The variation in bacterial response to AgNPs could be attributed to the nature and structure of bacterial cells, their resistance mechanisms, and/or the surface chemistry of AgNPs, which have a high affinity for bacterial membranes. Our biogenic AgNPs exhibited biological activities comparable to or greater than those previously reported (Table S3).

Overall, these findings highlight that the biological performance of biogenic AgNPs is governed not solely by particle-size-dependent reactivity but also by corona-mediated cellular interactions, offering a rational basis for the design of sustainable nanomaterials for targeted biomedical applications.

Conclusion

In this study, we developed a sustainable and eco-friendly synthesis method to produce well-dispersed, small AgNPs using cell-free medium and biomass extracts of PCC 6307 as reducing and stabilizing agents, without the use of additional surfactants. Reaction conditions were systematically optimized to control the synthesis of small, stable AgNPs. The optimized reaction produced monodisperse CS@AgNPs and CB@AgNPs with predominantly spherical particles and average diameters of 4.0 nm and 12.4 nm, respectively. UV-Vis spectroscopy, TEM, SEM, EDx, Zetasizer measurements, and FTIR confirmed the successful formation of stable and small AgNPs coated with an algal corona. These coronas facilitate nanoparticle stability and functionality. These naturally functionalized AgNPs exhibited notable biological activities, with CS@AgNPs and CB@AgNPs showing significant anticancer activity against the tested colon cancer cell lines.

Biogenic AgNPs exhibited moderate radical scavenging activity, which may be attributed to algal corona coating their surfaces. The biogenic AgNPs showed stronger biocidal activity than chemically synthesized AgNPs. Mechanistically, the enhanced functionality of biogenic AgNPs is likely due to the synergistic effect of Ag⁺ ion reactivity and the algal corona, which enhances nanoparticle stability and promotes NP/cell interactions. Overall, this study highlights *C. gracile* PCC 6307 as a promising biofactory for the synthesis of uniform, small spherical AgNPs with promising biological activities. Further metabolomic and proteomic investigations are required to better understand the synthesis mechanism of AgNPs and further improve their biological activities.

Data availability

The data supporting this article are available in Figs. 1–7, supplementary Figures 1–4, Tables 1 and 2, and supplementary Tables 1–3. The algal specimens and datasets analyzed in the present study are available from the corresponding author upon reasonable request. The 16S rRNA sequence of **C. gracile** PCC 6307 is available in GenBank (accession number: OP360943.1).

Received: 22 July 2025; Accepted: 22 May 2026

Published online: 27 July 2026

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Acknowledgements

The authors would like to acknowledge Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R84), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

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R.S.H., M.A.A., M.B.A., H.A., Z.N.A., M.I.A., M.A.M., W.A.O., and S.M.A. contributed to the conception and design of the study. R.S.H., M.A.A., M.B.A., and M.M.A. were responsible for methodology, downloading and organizing datasets. R.S.H. performed the statistical and result analysis and wrote the first draft of the manuscript. All authors contributed to manuscript revision, read, and approved the submitted version.

Funding

The authors would like to acknowledge Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R84), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-55256-8>.

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